11/02/2016

Assignment No 1 Optical fiber Communication

Q1.) A cylindrical waveguide is made with film refractive index of 3.5 and cladding refractive index of 3.0. Determine the maximum thickness of film that will allow no more than six guided modes to propagate for $\lambda = 1.5\ \mu m$.

Q2.) For a multimode step index fiber the refractive indices of core and cladding are 1.53 and 1.48 respectively. Find out the angle of acceptance and critical angle when light is incident from air.

Q3.) For a step index fiber the normalized frequency (V) is 26.6 at a wavelength of 1300 nm. Determine the numerical Aperture if the core radius is 25 μm.

Q4.) Determine the numerical aperture of a single mode fiber where $n_1 = 1.4675$ and $n_2 = 1.4622$.

Q5) For a step index fiber the following properties are given:

$a = 4\ \mu m$, $n_1 = 1.5$, $n_2 = 1.47$.

Determine the cut off wavelength of the fiber.

Q6.) An optical fiber for single mode is to be designed for 1.55 μm. Communication system. Determine the maximum diameter of the core which can be allowed if $n_1 = 1.55$ and $n_2 = 1.48$.

Q7.) An optical fiber is operating at 0.84 μm wavelength. Determine the number of modes propagating through fiber if $a = 25\ \mu m$, $n_1 = 1.48$ and $n_2 = 0.01$.

Q8.) Determine the number of modes for a multimode fiber, if $a = 20\ \mu m$, $\lambda = 0.85\ \mu m$, $n_1 = 1.475$ and $n_2 = 1.472$. How many modes will propagate if λ is changed to 1.3 μm.

Q9.) A step index fiber is operating at a wavelength of 1.3 μm with the following parameters: $n_1 = 1.450$, $n_2 = 1.447$.

Determine: (a) Core radius; (b) NA; (c) Maximum acceptable angle